



Salt to freshwater in minutes?

A team of Korean researchers have made a membrane that can turn saltwater into freshwater. <https://digit.in/jul21-72>



Artificial meat finally a reality?

A newly launched facility in Israel has to generate 500 kgs of cultured products each day. <https://digit.in/jul21-73>

ment of these rocks. But, as history has it, there hasn't ever been an easy route to decoding such strange phenomenon. And these two people failed in drawing any conclusion too. They conducted a series of experiments by placing rocks on the bed of the Racetrack Playa, but the movement patterns of these rocks were not able to provide any substantial and definitive evidence which could prove or give rise to any reasoning behind why these rocks had been exhibiting such a strange behaviour.

Efforts continued but, as we told you earlier, it wasn't until as recent as 2014, that the mystery was finally solved. And the men who deserve the credit for uncovering the secrets of the moving stones were a part of a research group that had been aided by the Scripps Institution of Oceanography, NASA and other such institutions, and were a part of an initiative which was named the Slithering Stones Research Initiative. At the forefront of this were two cousins who go by the name, Richard



Richard Norris, one of the researchers in Slithering Stones Research Initiative

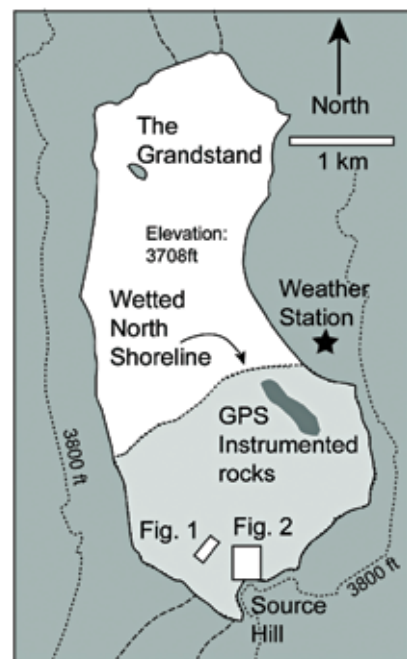
Source: San Diego State University

D. and James M. Norris, who had set out to dig further into this well of uncertainty and did finally hit the holy grail after about 3 years of meticulous study. They had set out to study these moving rocks in 2011 and were the ones that gave a name to their mission, which got several members attached to it.

As a part of their study, they planted 15 of their own stones on the surface of the Racetrack Playa and set up a weather station nearby. One of their masterstrokes was that they added GPS trackers to these rocks, which helped them monitor any movement that these rocks made, even if it was spread over such a long period of time that any camera or recording equipment would have failed to run continuously. They waited and waited and waited, until in December 2013, they got their first major breakthrough.

On the 4th and 20th of December, their setup, which had included time-lapse cameras, caught the rocks sliding across the playa at the rate of about 3 to 5 meters per minute. This was accompanied by several other such sightings through to January 2014. The rocks would move for about five minutes after no movement for months, and at times for sixteen minutes continuously. What they observed they jotted down in a research paper, which was published later and helped decode the mystery behind the sailing stones. An extract from the paper reads, "all authors agree that rocks are most likely to move when the playa surface is wet, creating a slick surface, and that wind must be involved" explains a lot. Upon further reading into the paper, we get to know that, what happened was that rains formed a small pool that froze overnight

and thawed the next day due to the heat from the sun. This resulted in the formation of large sheets of ice, which sometimes did not melt completely and floated on the water that had formed below them. When the wind blew, these ice sheets were pushed forward and gave us what we know was the sailing rocks phenomenon. One of the two researchers whom we had talked about earlier, in his statement regarding his group's discovery, attributed much of this early success to sheer luck and nothing else. He said, "Science sometimes has an element of luck. We expected to wait five or ten years without anything moving, but only two years into the project, we just happened to be there at the right



Map of Racetrack Playa

Source: Research Article titled Sliding Rocks on Racetrack Playa, Death Valley National Park: First Observation of Rocks in Motion on PLUS ONE



A rock with a GPS unit fixed inside it

time to see it happen in person." For almost a century this puzzling phenomenon was called different things, sailing rocks, moving rocks, walking rocks, and our favourite of them all, rolling stones, was actually just force of nature showing its raw presence on planet earth and giving rise to something that still remains to be something spectacular to observe first hand, or even in pictures. 